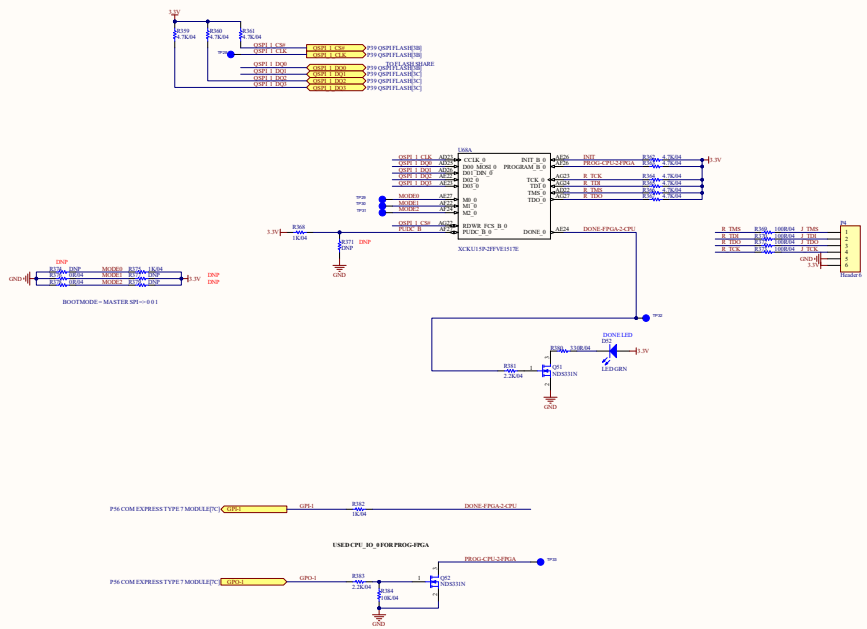
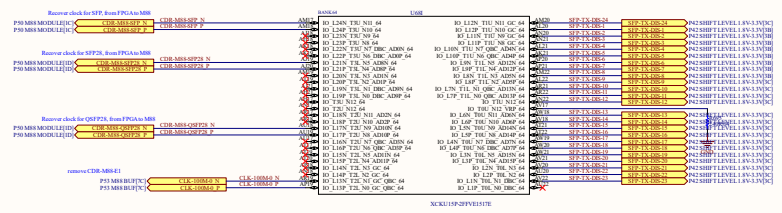






project: N3-HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi	
Size: A4	Number: 10	Revision: V1		
Date: 10/1/2024	Time: 8:45:18 AM	Sheet 10 of 71		
File: P10 BANK 64.SchDoc				

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Hanoi

The entire read data has must be placed in a single bank irrespective of the number of memory components. ok

All byte lanes that are used for the read data of a single component must be adjacent, no skip byte lanes are allowed. ok

On a different data type of memory component should be allocated in the same byte lanes (byte lanes 1 and 2). ok

If a byte lanes used for read data, then it can't be used for write. Read/Write (CQ) or CQP) give the first priority and data (D) in the next. ok

Read data lanes of two components should not share byte lanes. ok

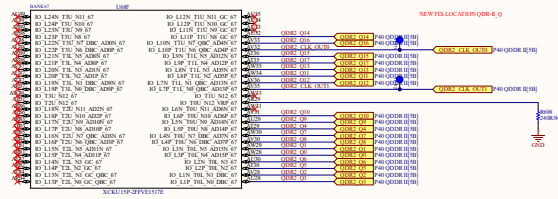
Read/Write pair must be allocated in one of the byte lanes that are used for the read. ok

CQ/CQP pair must be allocated in a single byte lane. ok

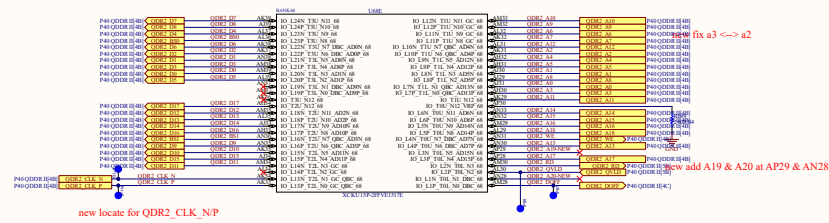
CQ/CQP must be allocated in the byte lanes 1 and 2 because the other type. ok

CQ/CQP must be allocated in the same byte lanes of all used byte lanes. If two byte lanes are used for the read data, either one of them can be used for CQ/CQP allocation. ok

CQ and CQP must be allocated in either pair or pair to the byte lanes. For example, if CQ is allocated to pair 0, CQP should be allocated to pair 1 and vice versa. ok



Band BW01 is same bank 0k
Only one write data bus is allowed per byte bank 0k
All byte lanes that are used for the write data of a single component must be adjacent, no skip byte lanes are allowed 0k
One of the write data bus is memory component should be allowed in the 0k
Memory Clock pair must be adjacent in the write data of the corresponding memory component. 0k
Memory clock should come from one of the write data bus (byte lanes 1 and 2). 0k
K/C can be located in any P/P pair 0k
All address control (A/C) bits must be located in a single bank 0k
All A/C bit lanes should be contiguous and no skip byte lanes is allowed 0k
The address control of bank should be the same or adjacent to that of the write data bank 0k
There should be no any byte lanes crossing between A/C and write data byte lanes. This rule applies when A/C and write data share the same bank or adjacent in adjacent bank 0k
Address control pair should not share a byte lane with the write data as well as vice 0k



project: **N3-HTPSR5G**

Size: **A4**

Number: **14**

Revision: **V1**

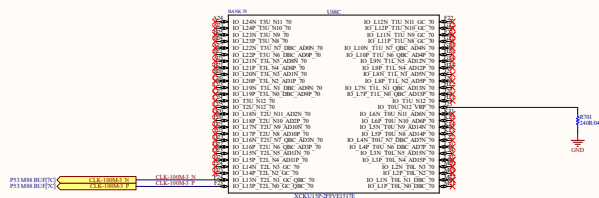
Date: **10/1/2024**

Time: **8:45:19 AM**

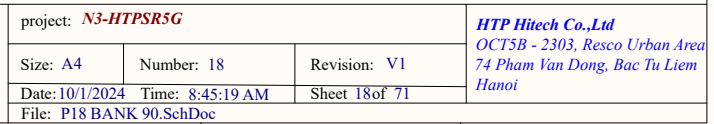
Sheet **14** of **71**

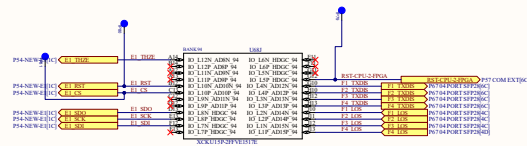
File: **P14 BANK 68.SchDoc**

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project: N3-HTPSR5G			HTP Hitech Co.,Ltd <i>OCT5B - 2303, Resco Urban Area</i> <i>74 Pham Van Dong, Bac Tu Liem</i> <i>Hanoi</i>
Size: A4	Number: 16	Revision: V1	
Date: 10/1/2024	Time: 8:45:19 AM	Sheet 16 of 71	
File: P16 BANK 70.SchDoc			



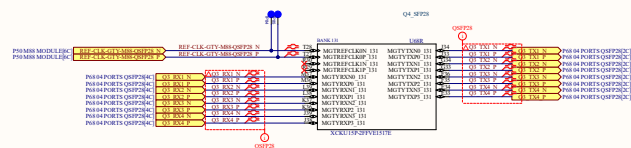


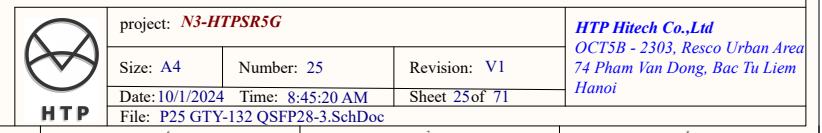
Output Pin, BE/SET FPGA from CPU

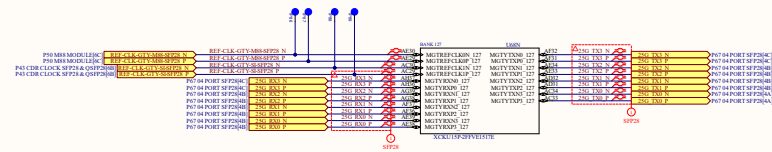


File: P23 GTY-130 QSFP28-1.SchDoc

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Hanoi

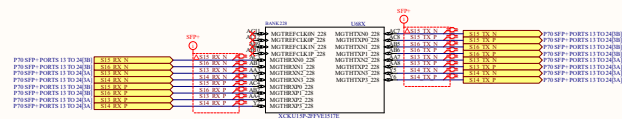






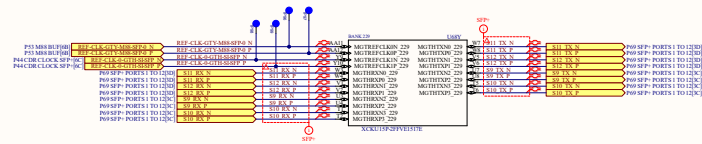


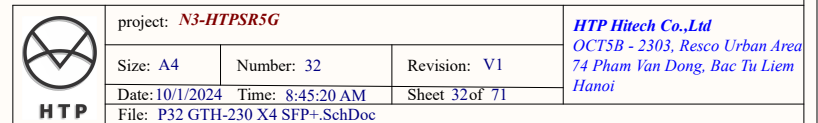
project: N3-HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi	
Size: A4	Number: 28	Revision: V1		
Date: 10/1/2024	Time: 8:45:20 AM	Sheet 28 of 71		
File: P28 GTH-224 X4 SFP+.SchDoc				

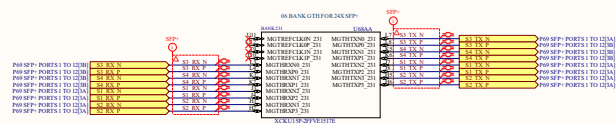


 HTP	project: <i>N3-HTPSR5G</i>			<i>HTP Hitech Co.,Ltd</i> <i>OCT5B - 2303, Resco Urban Area</i> <i>74 Pham Van Dong, Bac Tu Liem</i> <i>Hanoi</i>
	Size: A4	Number: 30	Revision: V1	
	Date: 10/1/2024	Time: 8:45:20 AM	Sheet 30 of 71	
	File: P30 GTH-228 X4 SFP+.SchDoc			

PLACE REF CLK HERE FOR SHARE 64 BANKS







```
01 BANK GTF FOR PCIE34          NEED CORRECT BY PRE-BUILD VIVADO PROJECT FOR PCIE
REF CLK FROM PCIE.CONN
```

[illegible]project: *N3-HTPSR5G*

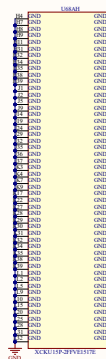
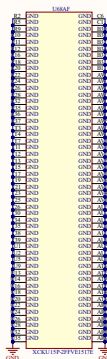
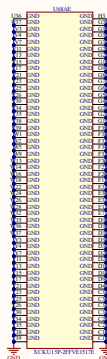
Size: A4

Number: 34

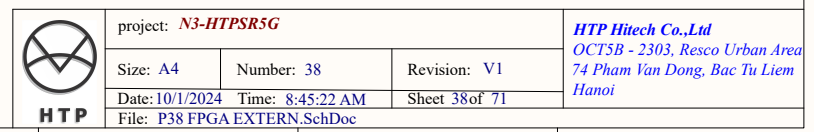
Revision: V1

Date: 10/1/2024 Time: 8:45:21 AM

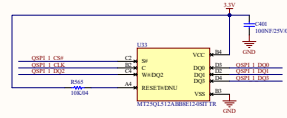
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Hanoi



ADD 02 GND CONNECT HERE, AF17 & L32



CHECK DUNG LUONG
12/2/24
Table 3-4: Minimum Length for UltraScale Architecture-based FPGAs
KUTOP3202: 28K, 74K, 90K, 160K, 200K, 300K, 400K, 550K, 700K, 850K, 100K



Pin

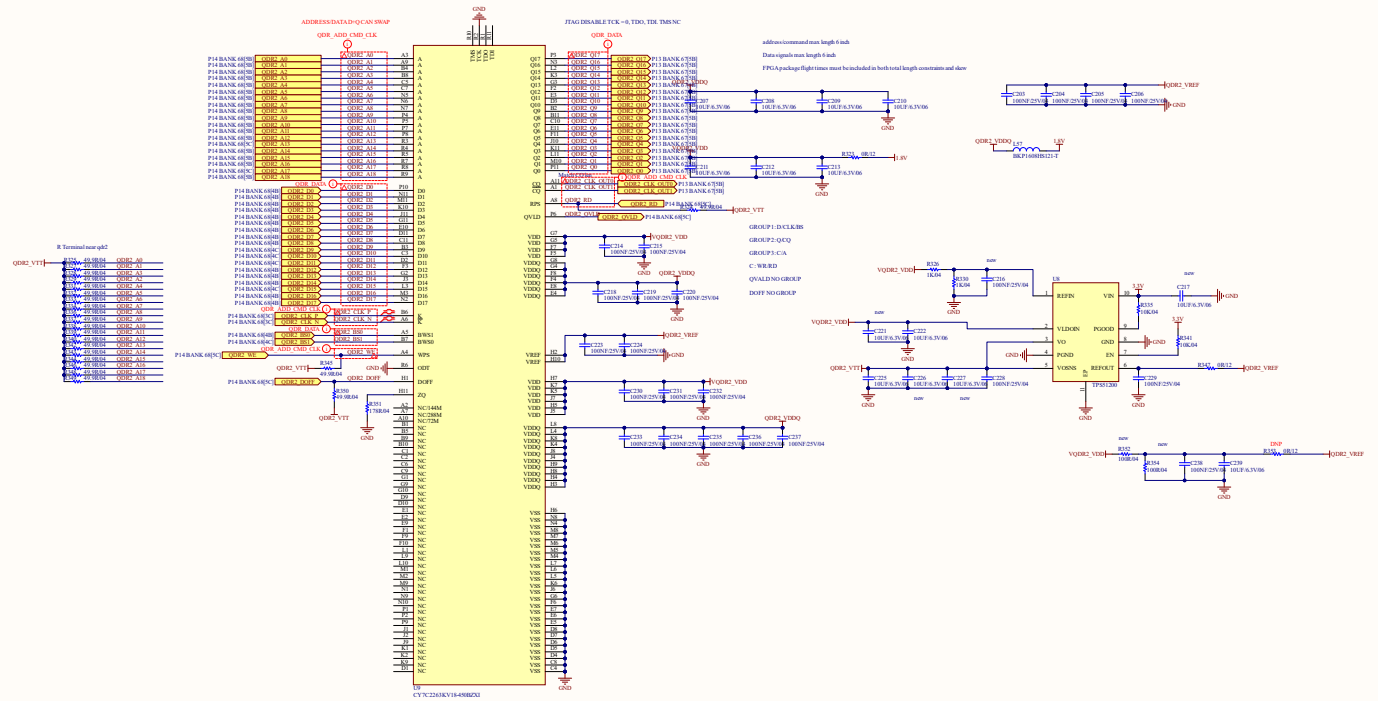
PCONFIG[0]	Q0[0:15]
PCONFIG[1]	Q1[0:15]
PCONFIG[2]	Q2[0:15]
PCONFIG[3]	Q3[0:15]
PCONFIG[4]	Q4[0:15]
PCONFIG[5]	Q5[0:15]
PCONFIG[6]	Q6[0:15]
PCONFIG[7]	Q7[0:15]



project: N3-HTPSR5G			HTP Hitech Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi	
Size: A4	Number: 39	Revision: V1		
Date: 10/1/2024	Time: 8:45:22 AM	Sheet 39 of 71		
File: P39 QSPI FLASH.SchDoc				

QDR (USED) PWR SEQUENCE
 1.VV_VDD 1.VV_VDD
 1.VV_VDDQ 1.VV_VDDQ
 QDR2_VTT QDR2_VTT

PWR Sequence: STEP0:DOFF LOW→STEP1:VDD→VDDQ→VREF



project: **N3-HTPSR5G**

Size: A4

Number: 40

Revision: V1

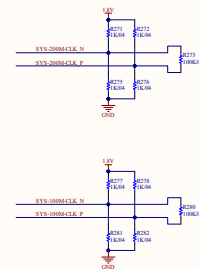
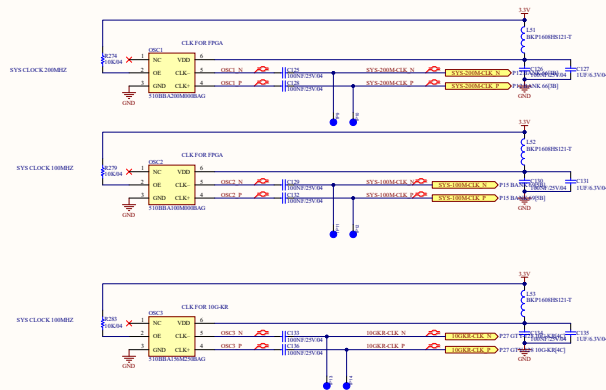
Date: 10/1/2024

Time: 8:45:22 AM

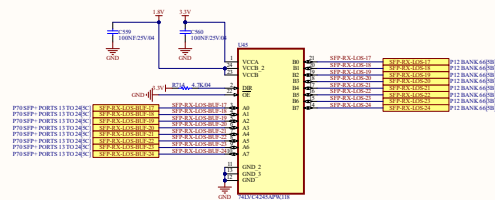
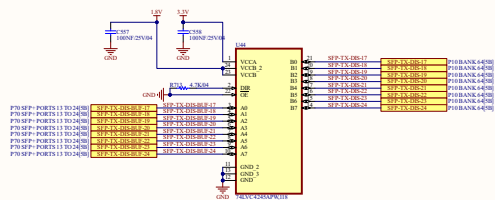
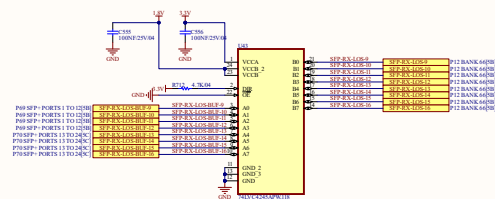
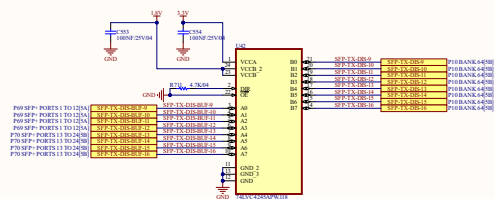
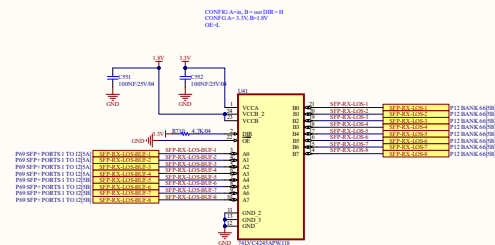
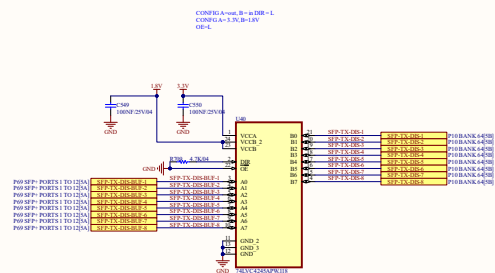
Sheet 40 of 71

File: P40 QDDR II.SchDoc

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 HTP	project: <i>N3-HTPSR5G</i>			<i>HTP Hitech Co.,Ltd</i> <i>OCT5B - 2303, Resco Urban Area</i> <i>74 Pham Van Dong, Bac Tu Liem</i> <i>Hanoi</i>
	Size: A4	Number: 41	Revision: V1	
	Date: 10/1/2024	Time: 8:45:22 AM	Sheet 41 of 71	
	File: P41 CLK FPGA OSC.SchDoc			

project: *N3-HTPSR5G*

Size: A4

Number: 42

Revision: V1

Date: 10/1/2024

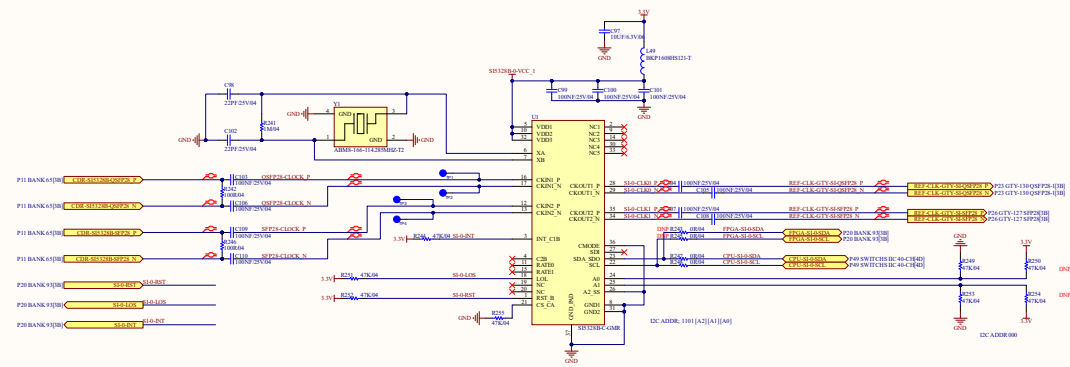
4 Time: 8:45:22 AM

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File: P42 SHIFT LEVEL 1.8V-3.3V.SchDoc

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Hanoi

OPTIONAL REF CLK/QFP28 TEST PURPOSE ONLY/NOT IMPLEMENT IN PRODUCT



project: **N3-HTPSR5G**

Size: A4

Number: 43

Revision: V1

Date: 10/1/2024

Time: 8:45:22 AM

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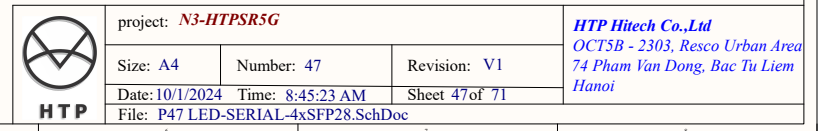
File: P43 CDR CLOCK SFP28 & QSFP28.SchDoc

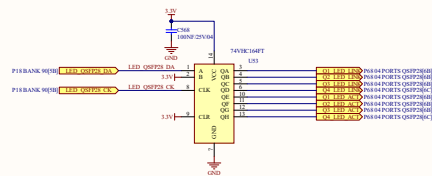
HTP Hitec Co.,Ltd
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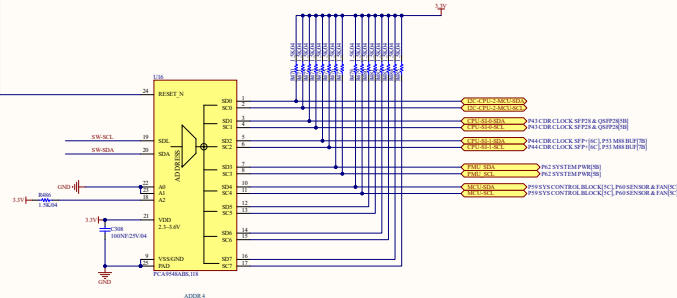
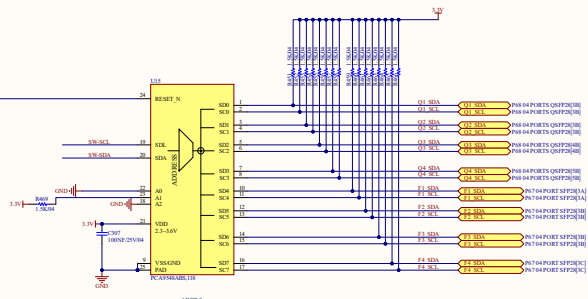
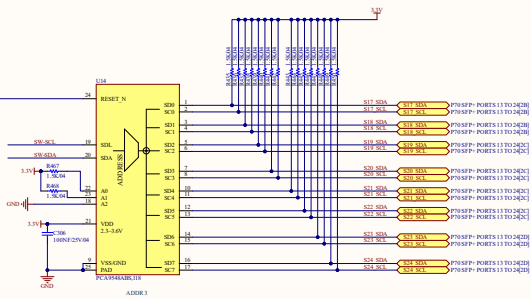
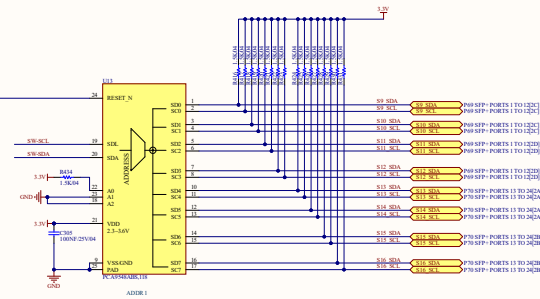
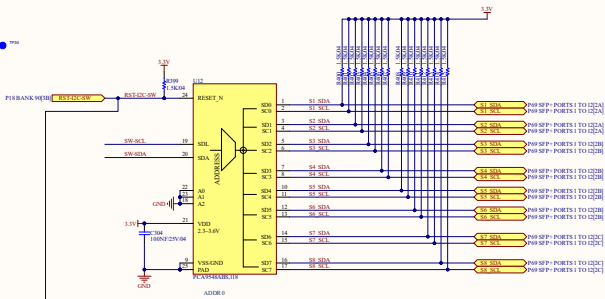


4 Time: 8:45:22 AM
R CLOCK SFP+.SchDoc

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project: **N3-HTPSR5G**

Size: A4

Number: 49

Revision: V1

Date: 10/1/2024

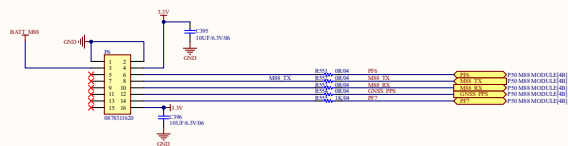
Time: 8:45:23 AM

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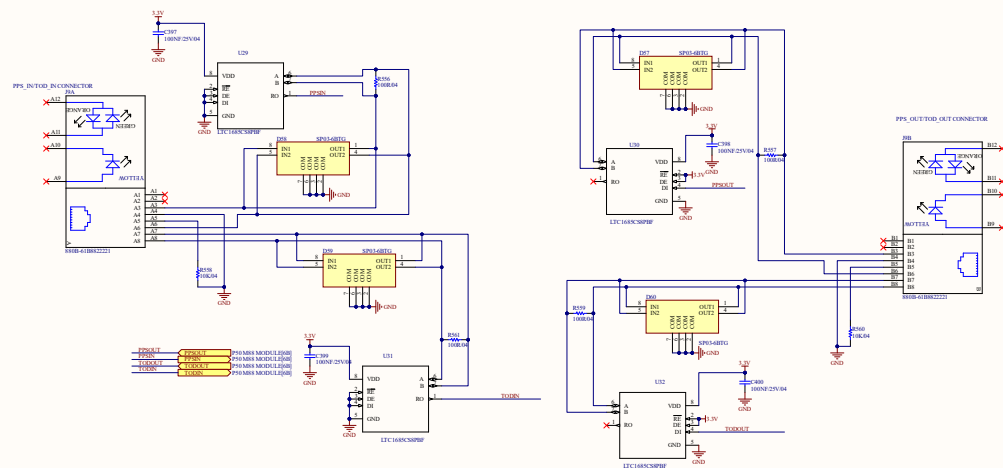
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Hanoi



 HTP	project: <i>N3-HTPSR5G</i>			<i>HTP Hitech Co.,Ltd</i> <i>OCT5B - 2303, Resco Urban Area</i> <i>74 Pham Van Dong, Bac Tu Liem</i> <i>Hanoi</i>
	Size: <i>A4</i>	Number: <i>51</i>	Revision: <i>V1</i>	
	Date: <i>10/1/2024</i>	Time: <i>8:45:23 AM</i>	Sheet <i>51</i> of <i>71</i>	
	File: <i>P51 M88 CONN GNSS.SchDoc</i>			



project: **N3-HTPSR5G**

Size: A4

Number: 52

Revision: V1

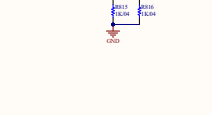
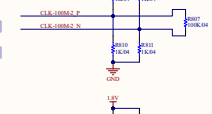
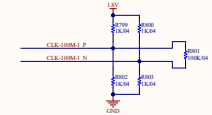
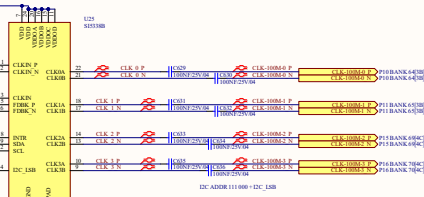
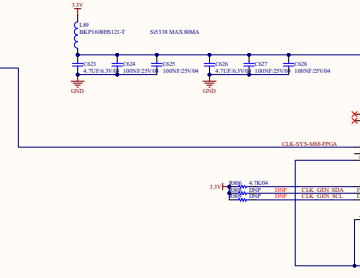
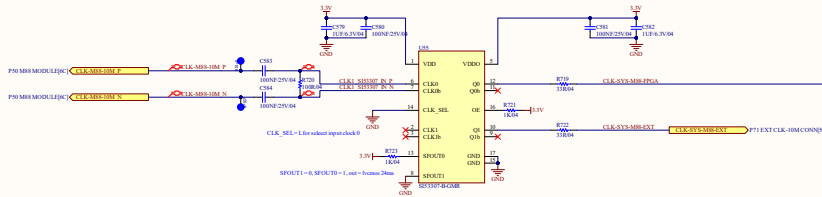
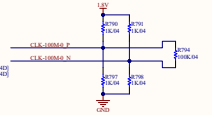
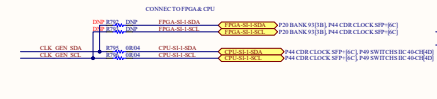
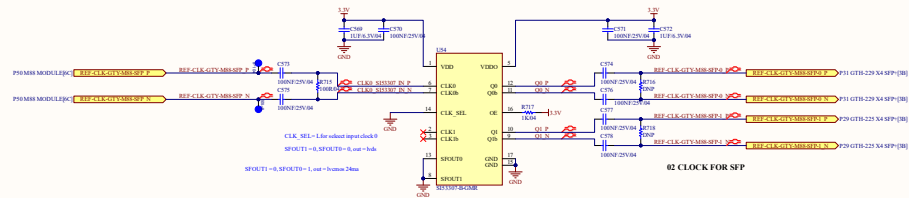
Date: 10/1/2024

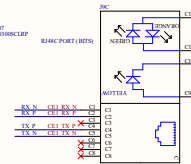
Time: 8:45:23 AM

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File: P52 M88 CONN PPS TOD.SchDoc

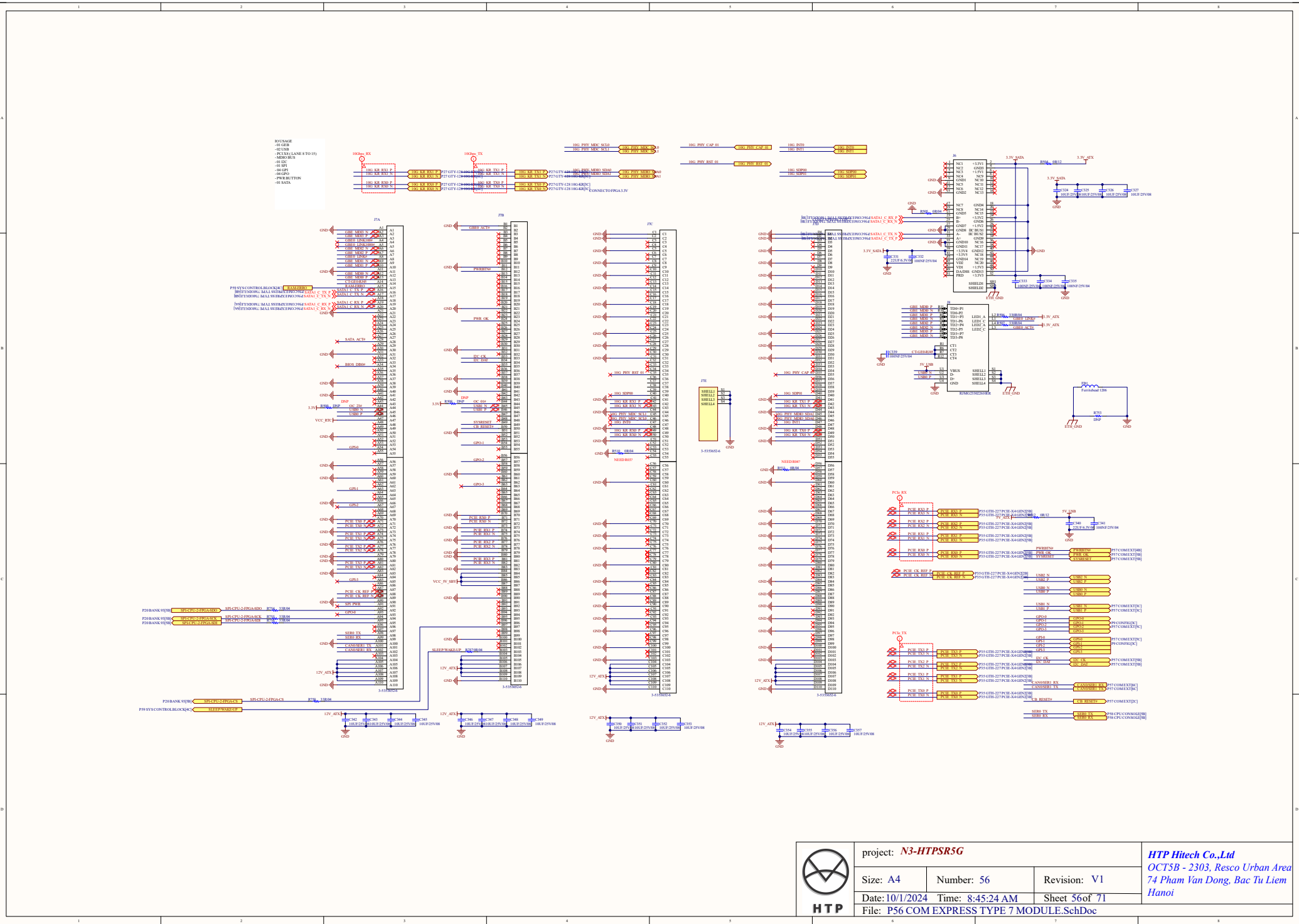
HTP Hitech Co.,Ltd
 OCT5B - 2303, Resco Urban Area
 74 Pham Van Dong, Bac Tu Liem
 Hanoi



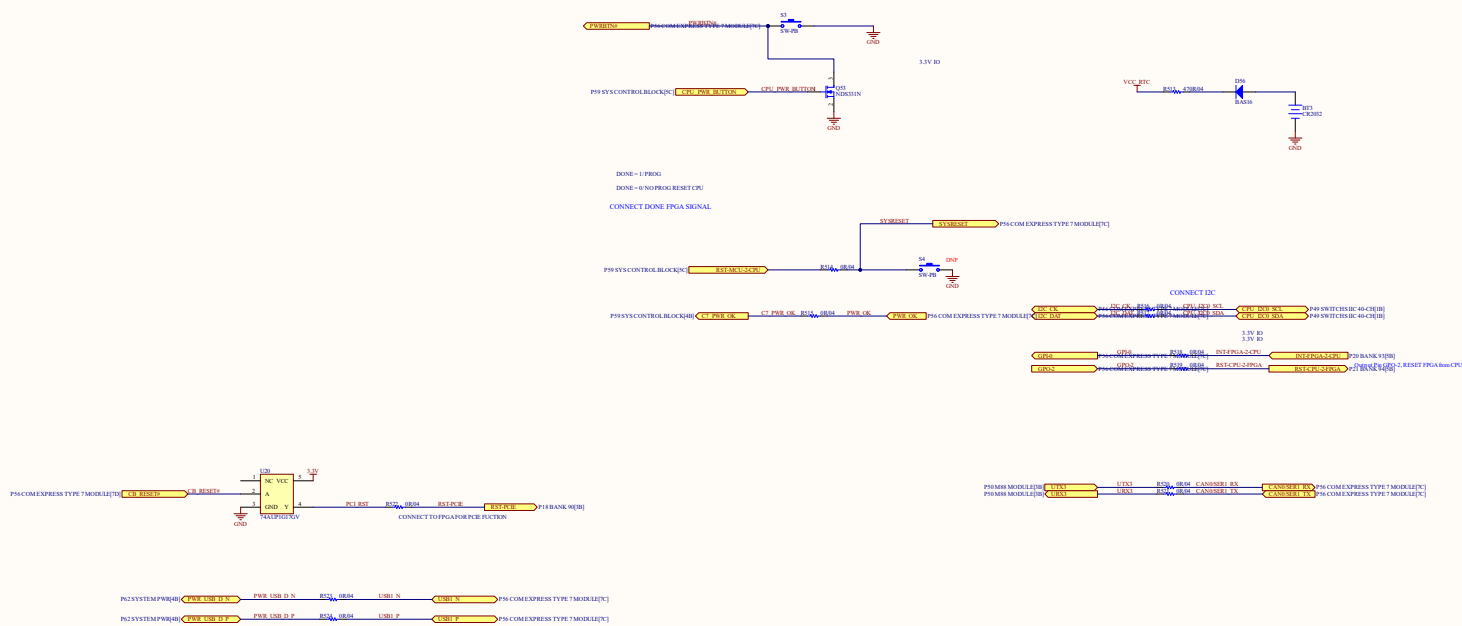


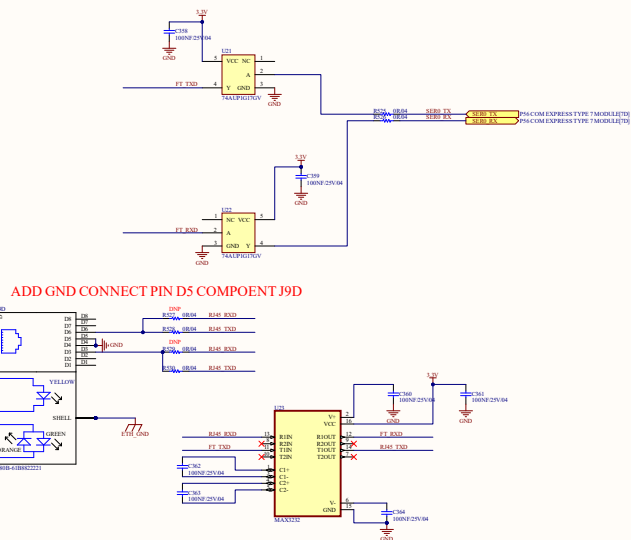
HTP Hitech Co.,Ltd
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Hanoi



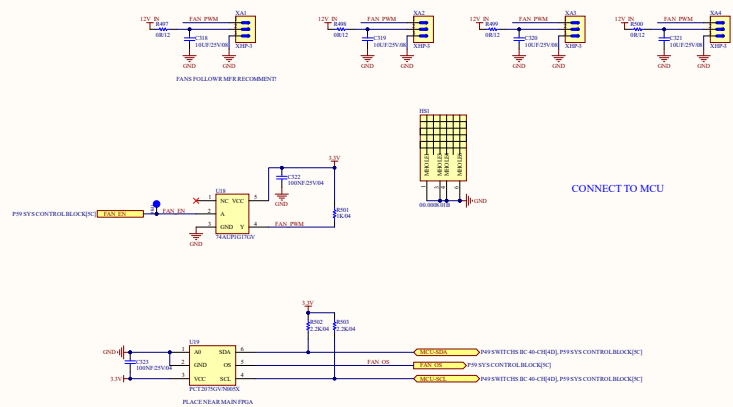


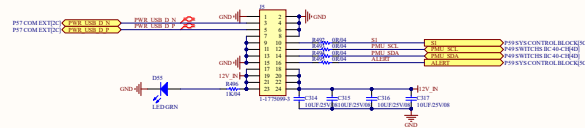
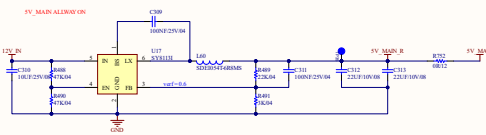
project: N3-HTPSR5G			HTP Hitech Co.,Ltd <i>OCT5B - 2303, Resco Urban Area</i> <i>74 Pham Van Dong, Bac Tu Liem</i> <i>Hanoi</i>
Size: A4	Number: 56	Revision: V1	
Date: 10/1/2024	Time: 8:45:24 AM	Sheet 56of 71	
File: P56 COM EXPRESS TYPE 7 MODULE.SchDoc			



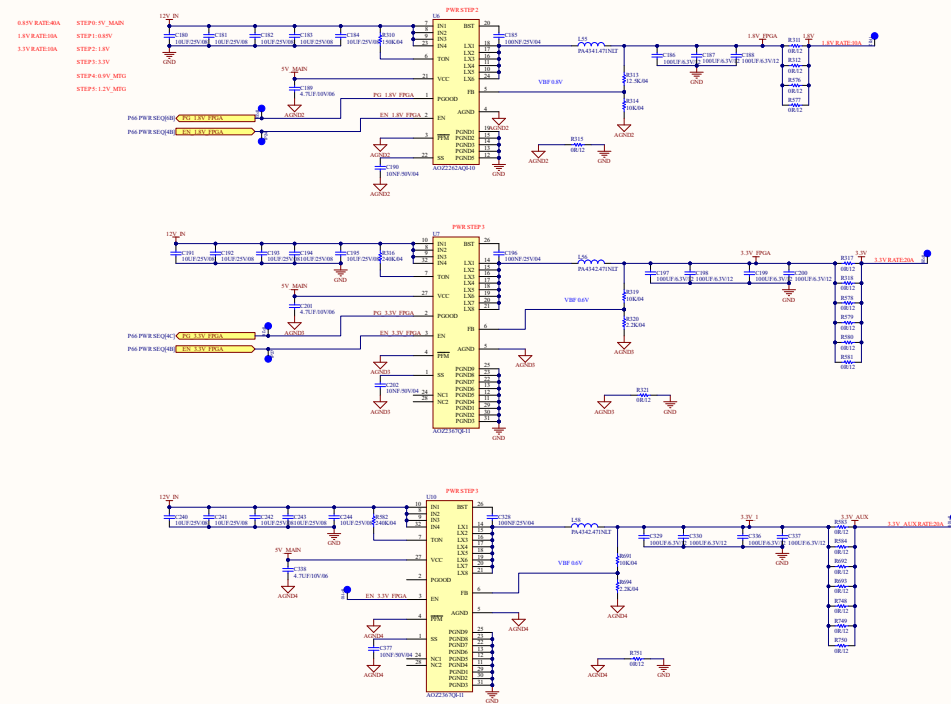


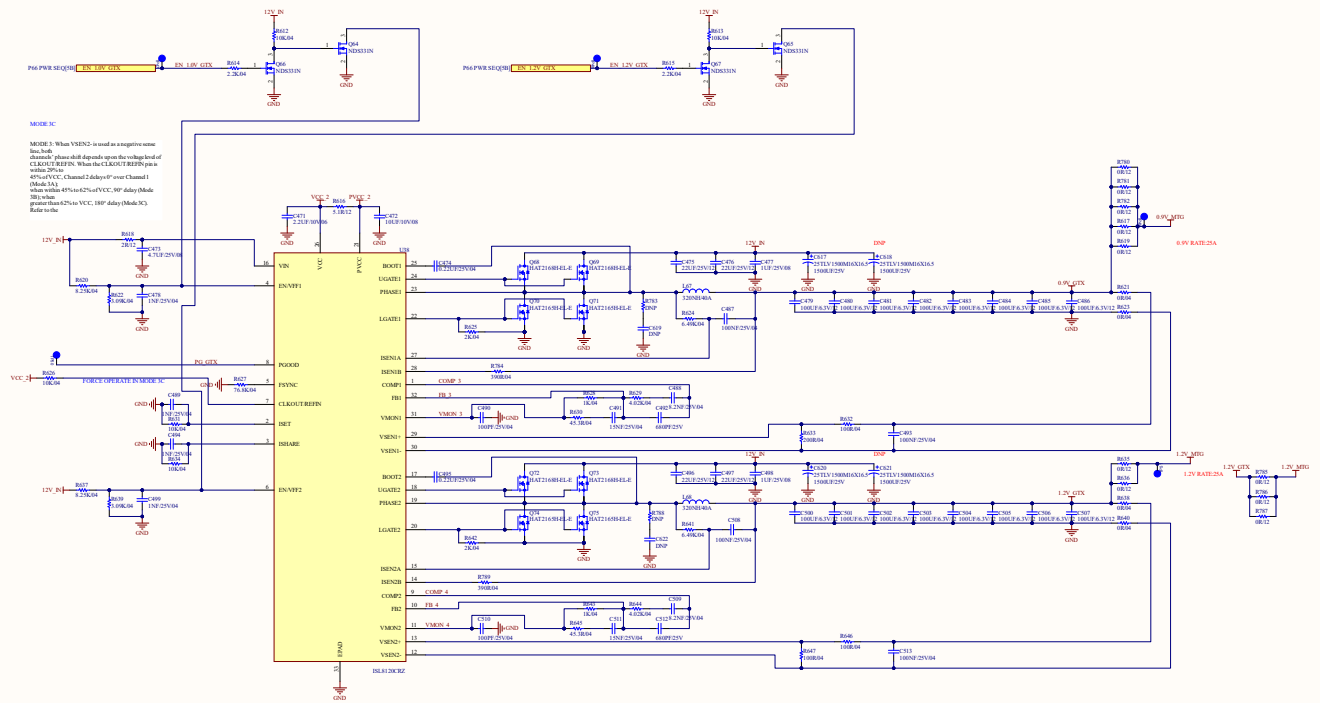
 HTP	project: N3-HTPSR5G			HTP Hitech Co.,Ltd	
	Size: A4	Number: 58	Revision: V1	OCT5B - 2303, Resco Urban Area	
	Date: 10/1/2024	Time: 8:45:24 AM	Sheet 58 of 71	74 Pham Van Dong, Bac Tu Liem	
	File: P58 CPU CONSOLE.SchDoc			Hanoi	



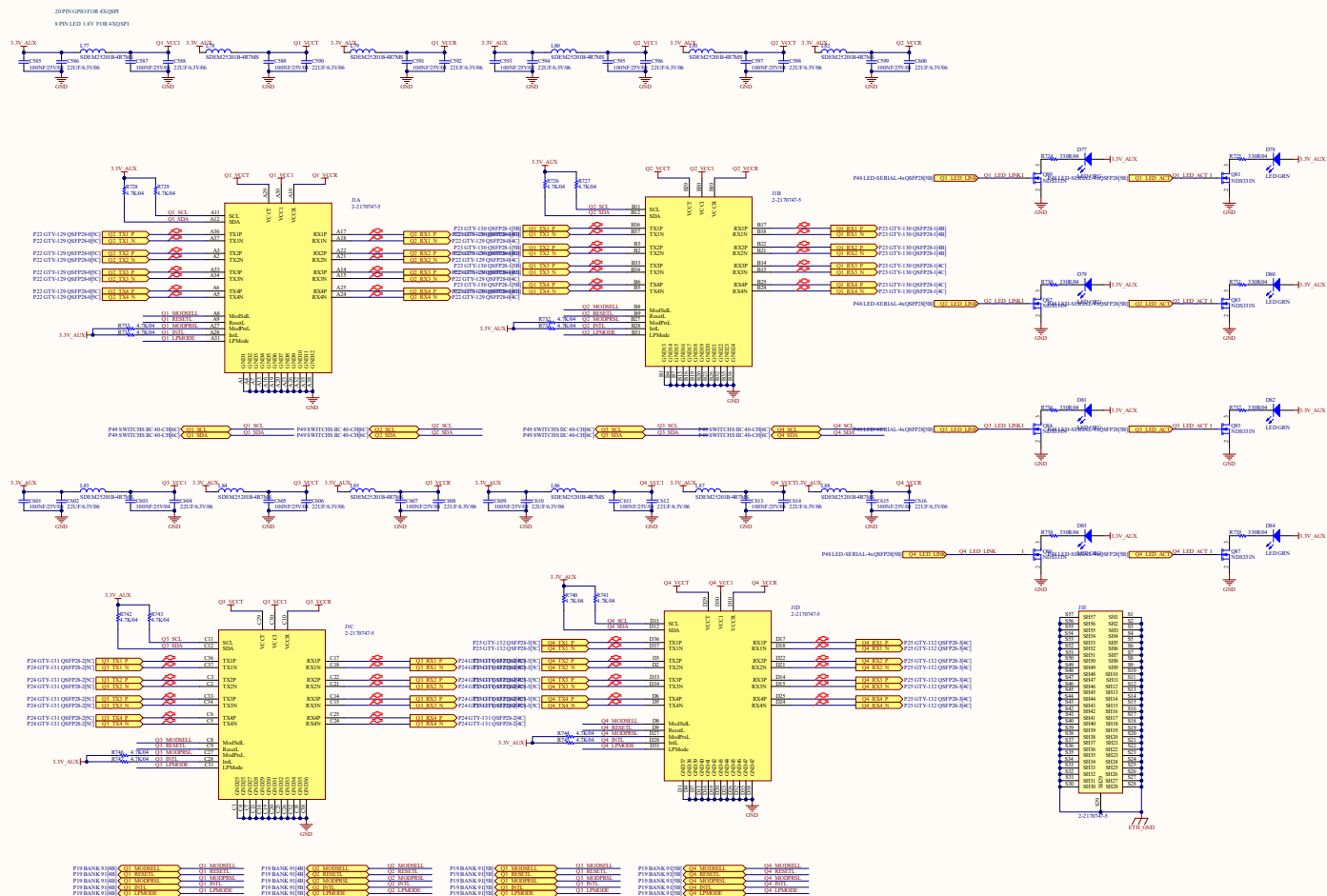


 HTP	project: <i>N3-HTPSR5G</i>			<i>HTP Hitech Co.,Ltd</i> <i>OCT5B - 2303, Resco Urban Area</i> <i>74 Pham Van Dong, Bac Tu Liem</i> <i>Hanoi</i>	
	Size: <i>A4</i>	Number: <i>62</i>	Revision: <i>V1</i>		
	Date: <i>10/1/2024</i>	Time: <i>8:45:25 AM</i>	Sheet <i>62</i> of <i>71</i>		
	File: <i>P62 SYSTEM PWR.SchDoc</i>				





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Hanoi



project: **N3-HTPSR5G**

Size: A4

Number: 68

Revision: V1

Date: 10/1/2024

Time: 8:45:26 AM

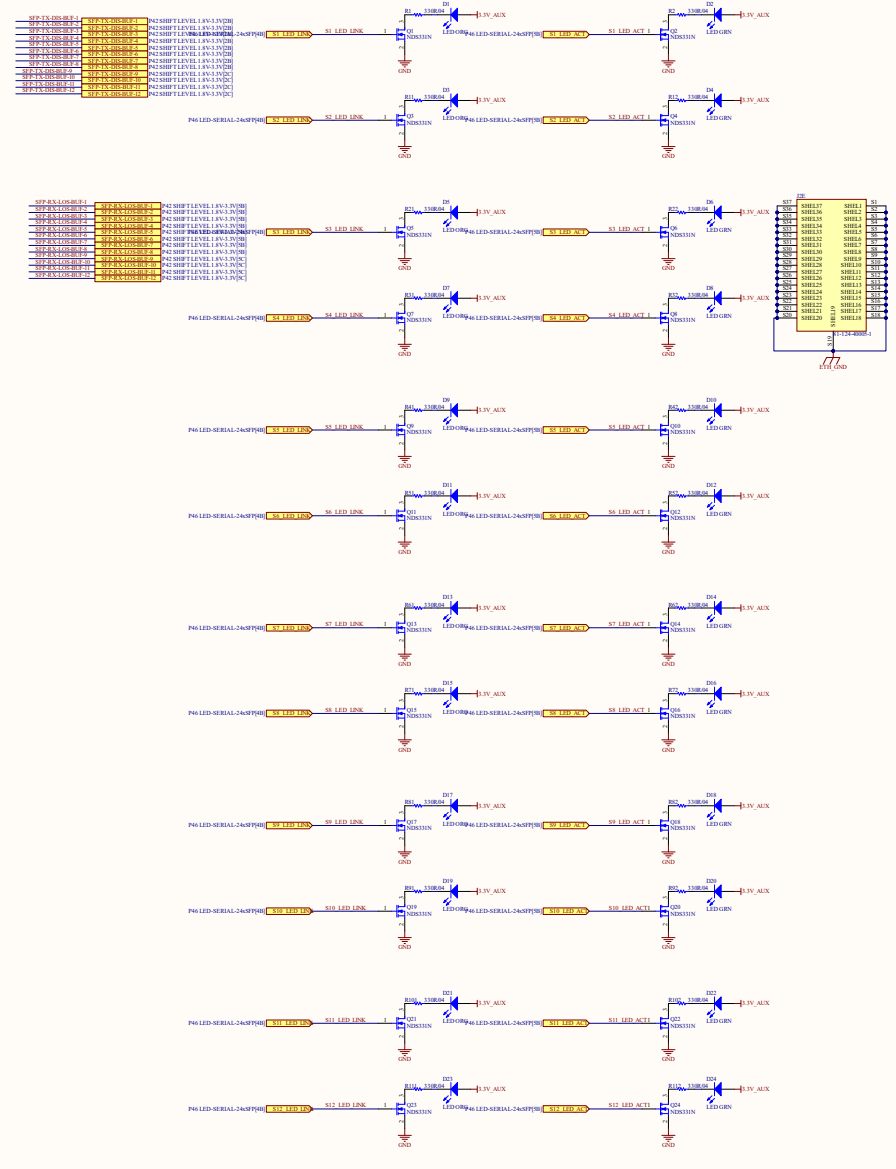
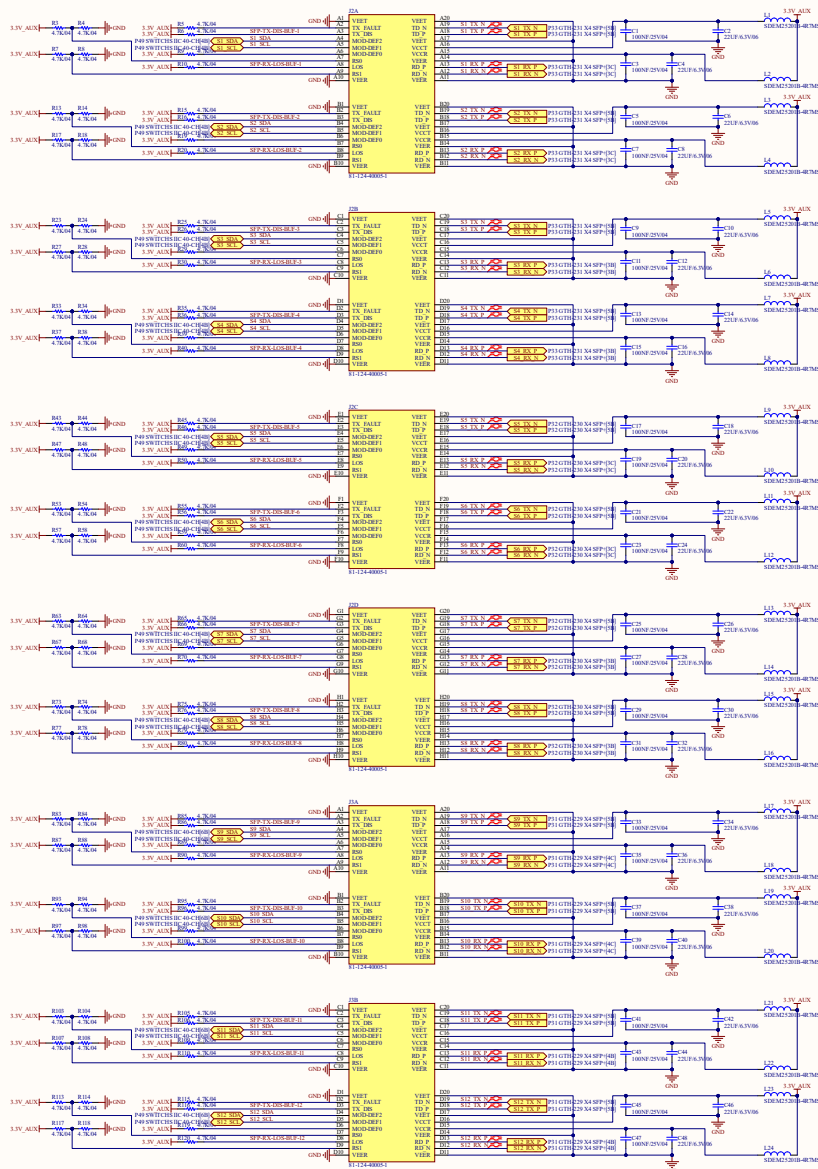
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Hanoi

EN62369-1 V1 FOR EX 1A
 EN62369-1 V1 FOR EX 1A-LED LINK-LED ACT
 EN62369-1 V1 FOR EX 1A-LED LINK-LED ACT
 EX 1A-LED LINK-LED ACT
 EX 1A-LED LINK-LED ACT

and 100V/1A act indicator 4.7k ohm 1A



project: **N3-HTPSR5G**

Size: A4

Number: 69

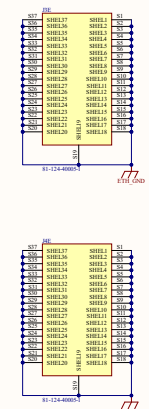
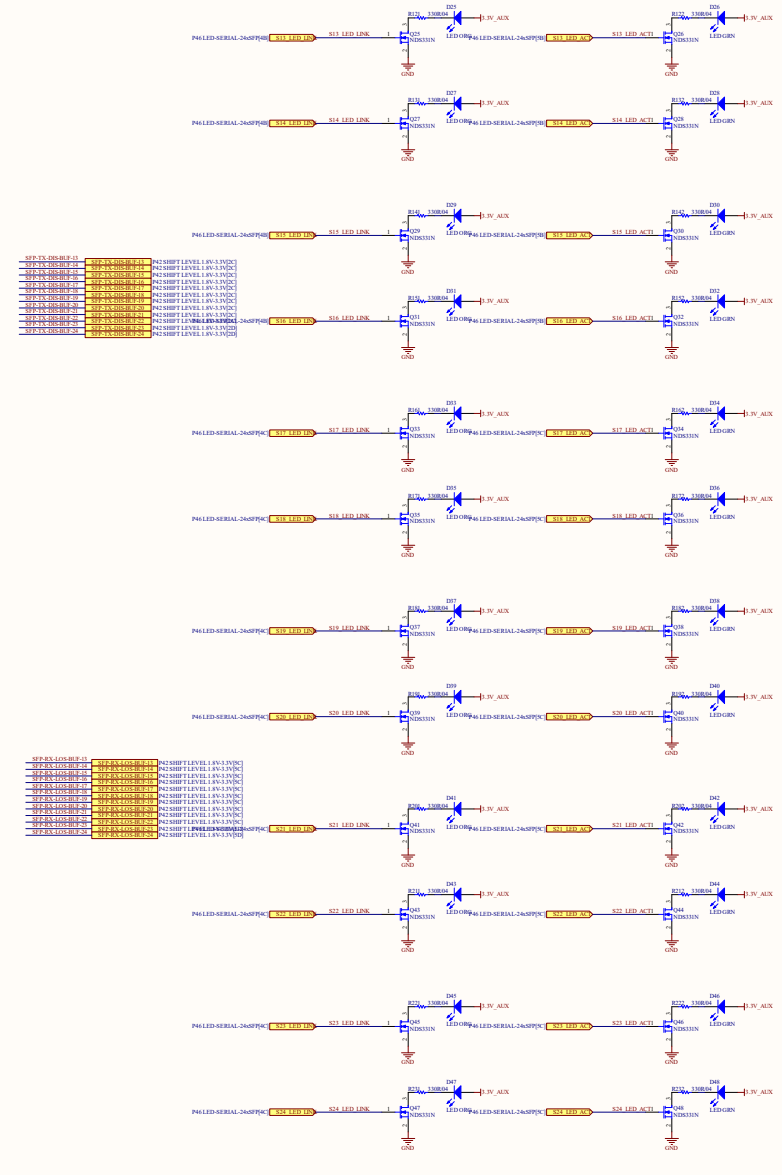
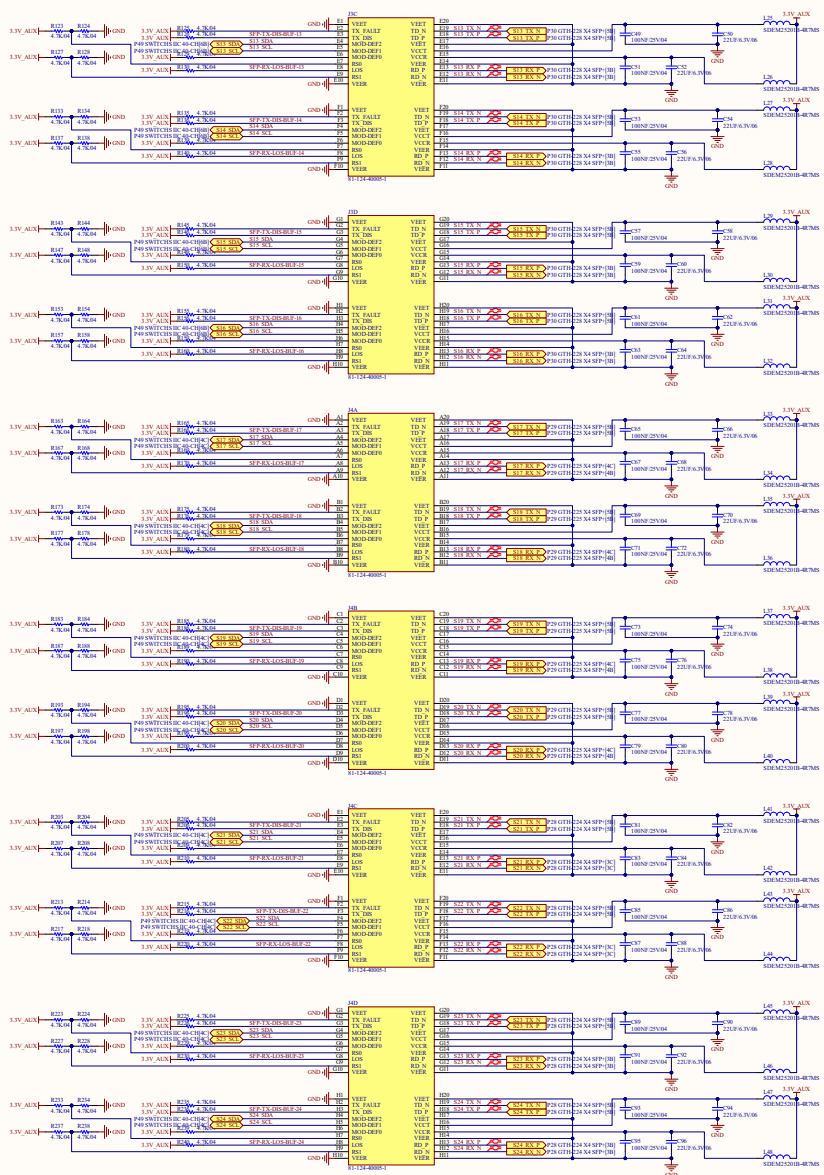
Revision: V1

Date: 10/1/2024 Time: 8:45:26 AM

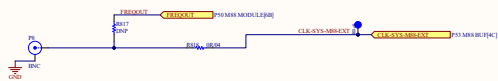
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File: P69 SFP+ PORTS 1 TO 12.SchDoc

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 HTP	project: N3-HTPSR5G		HTP Hitec Co.,Ltd OCT5B - 2303, Resco Urban Area 74 Pham Van Dong, Bac Tu Liem Hanoi
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 HTP	project: N3-HTPSR5G			HTP Hitech Co.,Ltd	
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	File: P71 EXT CLK-10M CONN.SchDoc			Hanoi	